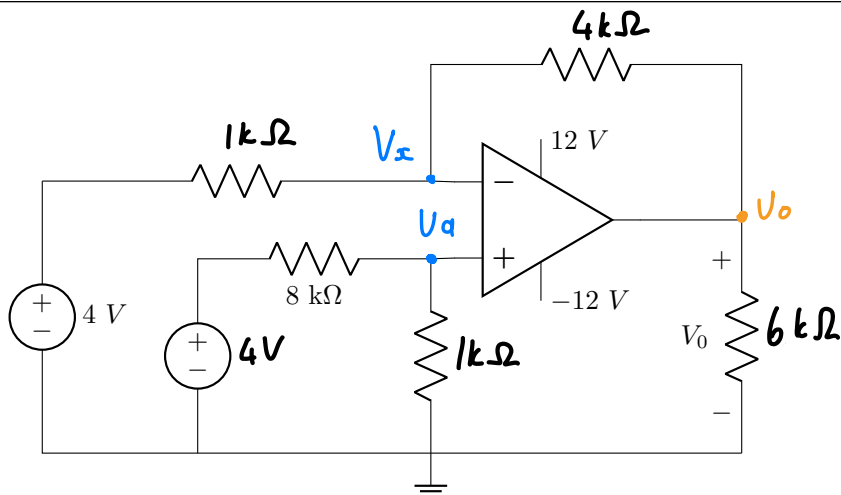


Section 1

Question 1 [2]

Given: V_a , R_1 , R_2 , R_3 , R_4 .01 Find V_0 [2]

@ V_a : $\frac{V_a - 4}{8000} + \frac{V_a}{1000} = 0$

$9V_a = 4$
 $V_a = 0.4444\text{ V}$

@ V_x : $\frac{V_x - 4}{1000} + \frac{V_x - V_0}{4000} = 0$

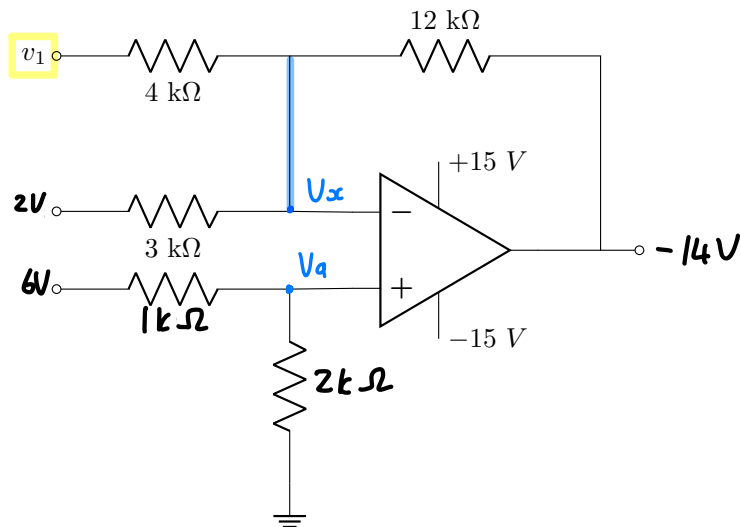
($V_a = V_x$ always)

$\frac{0.4444 - 4}{1000} + \frac{0.4444 - V_0}{4000} = 0$

$-14.224 + 0.4444 - V_0 = 0$

$V_0 = -13.7778\text{ V}$

Question 2 [2]

Given: v_0 , v_2 , v_3 , R_1 , R_2 .02 Find v_1 . [2]

$$\textcircled{a} V_a: \frac{V_a - 6}{1000} + \frac{V_a}{2000} = 0$$

$$3V_a = 12$$

$$V_a = 4V$$

$$\textcircled{b} V_x: \frac{V_x - 2}{3000} + \frac{V_x - V_1}{4000} + \frac{V_x + 14}{12000} = 0$$

$$4V_x - 8 + 3V_x - 3V_1 + V_x + 14 = 0$$

$$16 - 8 + 12 - 3V_1 + 4 + 14 = 0$$

$$3V_1 = 38$$

$$V_1 = 12.667V$$

Questions 3 to 5 [4]	
Given: V_{in} .	
03	Determine V_{o1} [1]
04	Determine V_{o2} [1]
Given: $V_{o2} = 3V$	
05	Determine V_o [2]

③ $V_{a1} = 0V = V_{x1}$

④ $V_{o2} = V_{o1} = 0.49V$

@ V_{x1} : $\frac{V_{x1} - 11 \times 10^{-3}}{10\,000} + \frac{V_{x1} + 120 \times 10^{-3}}{20\,000} + \frac{V_{x1}}{25\,000} + \frac{V_{x1} - V_{o1}}{100\,000} = 0$

$-0.11 + 0.6 + 0 + 0 - V_{o1} = 0$

$V_{o1} = 0.49V$

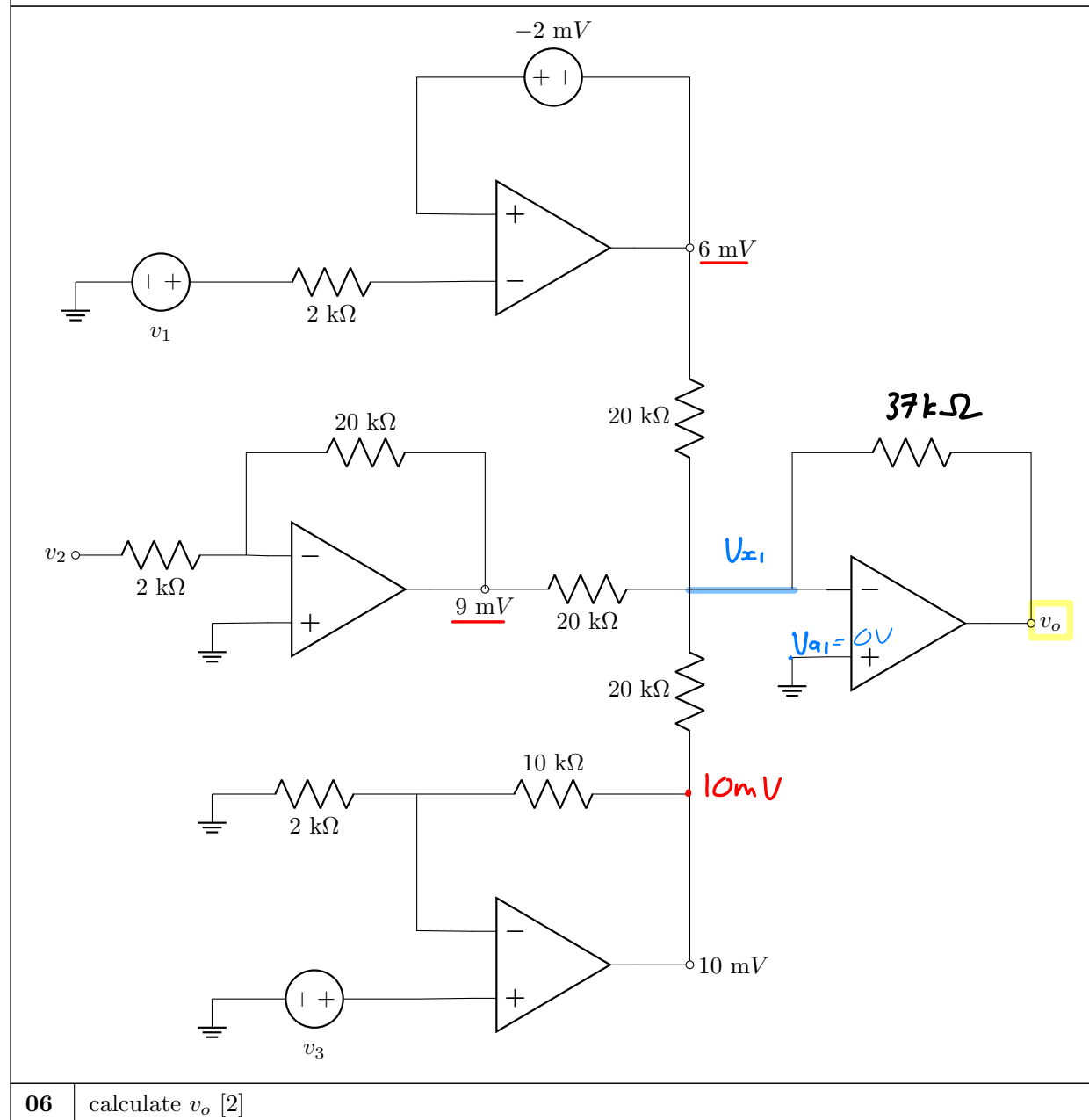
⑤ $V_{a3} = V_{o2} = 3V$

@ V_{a3} : $\frac{V_{a3} + 5}{200\,000} + \frac{V_{a3} - V_o}{20\,000} = 0$

$3 + 5 + 30 - 10V_o = 0$

$V_o = 3.8V$

Question 6 [2]

Given: R .06 calculate v_o [2]

$$\begin{aligned}
 &V_{a1} = 0V = V_{x1} \\
 &\text{@ } V_{x1}: \frac{V_{x1} - 0 \times 10^{-3}}{20\,000} + \frac{V_{x1} - 6 \times 10^{-3}}{20\,000} + \frac{V_{x1} - 10 \times 10^{-3}}{20\,000} + \frac{V_{x1} - v_o}{37\,000} = 0 \\
 &-16.65\text{ mV} - 11.1\text{ mV} - 18.5\text{ mV} - v_o = 0 \\
 &\underline{v_o = -46.25\text{ mV}}
 \end{aligned}$$

Total Section 1: [10]

Grand Total : [10]